

Chengyuan Luo

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Pittsburgh, USA

Aug 2026 – Aug 2028 (Expected)

Shanghai Jiao Tong University

Bachelor's degree in French, minor in Information Engineering

Shanghai, China

Aug 2021 – Jun 2025

GPA: 4.12/4.3 (Minor), 3.98/4.3 (Overall)

Selected coursework: Machine Learning (98/100), Mathematical Foundation for Artificial Intelligence (98/100), Data Structure (98/100), Probability and Statistics (98/100), Digital Signal Processing (98/100)

Honors and Awards:

- **Outstanding Graduate** of Shanghai Jiao Tong University, June 2025
- **Academic Scholarship (First Prize)** of SJTU-Paris Elite Institute of Technology (1/46), November 2024
- **Dean's Scholarship** of SJTU-Paris Elite Institute of Technology (1/73), September 2023
- **Gold Medal** in 46th International Collegiate Programming Contest (ICPC) Asia Regional Contest - Shanghai Site (rank 13/632) and Nanjing Site (rank 15/641), November and December 2021
- **Gold Medal** in 7th China Collegiate Programming Contest (CCPC) Weihai Site (rank 7/240), November 2021
- **Gold Medal** in China Computer Federation National Olympiad in Informatics Winter Camp (rank 18), August 2020

PUBLICATIONS

GBC: Generalized Behavior-Cloning Framework for Whole-Body Humanoid Imitation

arXiv 2025

Yifei He, **Chengyuan Luo**, Jiaheng Du, Wentao He, Jun-Guo Lu

AnyPlace: Learning Generalized Object Placement for Robot Manipulation

CoRL 2025

Yuchi Zhao, Miroslav Bogdanovic, **Chengyuan Luo**, Steven Tohme, Kouros Darvish, Alán Aspuru-Guzik, Florian Shkurti, Animesh Garg

RESEARCH EXPERIENCE

AGIBOT Innovation (Shanghai) Technology, Embodied VLA Algorithm Intern

Mar 2026 – Jul 2026

Project 1: Synthetic Data Generation for VLA Policy Training

- ▷ Built a synthetic demonstration workflow for task-specific VLA policy training.
- Implemented simulation teleoperation to record expert demonstrations and validate task execution.
- Developed mimicgen-style trajectory generation with task-tailored trajectory randomization.
- Added lighting and background randomization to improve visual robustness and generalization.
- ▷ Expanded limited expert demonstrations into diverse multimodal VLA training data.

Project 2: Pico VR Teleoperation for Humanoid Robots

- ▷ Developed Pico VR teleoperation for intervention and task execution on AGIBOT's latest humanoid robot.
- Mapped controller buttons and axes to robot actions, including dexterous hand control and intervention.
- Tuned inverse kinematics parameters for task-specific manipulation and intervention behavior.
- Built a monitoring interface to track robot status and manage control and recording modules.
- ▷ Enabled intervention workflows for real-world dexterous manipulation on a specific task.

ByteDance Seed Robotics, Intern

Oct 2025 – Feb 2026

Project: Large-Scale Data Generation for VLA Training

- ▷ Synthetic data generation for training VLA policies and evaluation of their performance in simulation.
- Developed general task setup enabling scalable configurations for new types of tasks.
- Optimized trajectory transform methods and improved inverse kinematics for tracking trajectories, increasing effective data generation efficiency by 48%.
- Improved time and memory efficiency of data generation simulation runs.
- ▷ Improved automated large-scale data generation pipelines that transformed a small number of human demonstrations into diverse, multimodal datasets optimized for VLA policies.

SJTU Machine Vision and Autonomous System Laboratory, Undergraduate Researcher *Dec 2024 – Jul 2025*
Supervised by Prof. Jun-Guo Lu, Shanghai Jiao Tong University

Project: Reinforcement Learning and Behavior Cloning for Bipedal Locomotion (GBC)

- ▷ Integration of behavior cloning into humanoid robot control policies.
- Pre-processed motion capture datasets to extract reference actions with additional information.
- Developed an efficient buffer to store reference actions with NVIDIA Warp during reinforcement learning.
- Modified the protocols of the reinforcement learning library `rs1_rl` for behavior cloning reward functions.
- Conducted extensive training and hyperparameter tuning, leading to improved policy performance.
- ▷ The article has been published on ArXiv.

People, AI, and Robotics (PAIR) Research Group, Undergraduate Researcher (online) *Aug 2024 – Feb 2025*
Advised by Prof. Animesh Garg, Georgia Institute of Technology.

Project: Object Placement Simulation for AnyPlace

- ▷ Implemented and simulated an object placement pipeline and evaluated the success rate.
- Modified the AnyGrasp model to generate grasps for diverse objects.
- Planned the pick-and-place trajectory using CuRobo to avoid collisions.
- Adapted the NVIDIA Isaac Lab grasping program for parallel executions of trajectories.
- Executed 20,000+ pick-and-place experiments across various objects and tasks.
- Analyzed the predicted placement poses and the simulated results to compute metrics for evaluation.
- ▷ The article has been accepted by **CoRL 2025**.

SJTU Machine Vision and Intelligence Group, Undergraduate Researcher *Feb 2023 – Dec 2024*
Advised by Prof. Cewu Lu, Shanghai Jiao Tong University.

Project: Benchmarking grasping models

- ▷ Implemented an automatic framework to evaluate 2-finger grasp models using multiple metrics.
- Designed a novel framework for 2-finger grasp models to test their performance.
- Developed the grasp simulation based on the framework in Bullet and NVIDIA Isaac Lab environment.
- Implemented an entire pipeline for calibration and testing grasps in the real world using ROS and MoveIt Motion Planning Framework and conducted extensive experiments.
- Wrote a program to control a microcontroller unit using FreeRTOS for multithreading.
- ▷ The framework can execute grasps automatically with little human intervention, and it can evaluate grasps comprehensively using multiple metrics.

WORK EXPERIENCE

ABB Engineering (Shanghai) Robot Research Lab, Research Intern for 3D Vision *Jun 2024 – Aug 2024*

Project: Object detection and pose estimation

- ▷ Identified objects and calculated their poses in a specific workspace.
- Implemented a framework to detect and estimate poses of specific objects using fiducial markers.
- Improved the detection using 2D object detection and segmentation models.
- Designed an algorithm to estimate poses using edge detection for objects with specific shapes.
- Complemented an additional academic survey on deep learning 3D reconstruction methods.
- ▷ Completed the internship's objectives with detailed documentation and several tests of the project.

EXTRACURRICULAR ACTIVITIES

SJTU RoboMaster Team, Team Member *Oct 2022 – Aug 2023*

- Improved the detection of opponents' robots based on YOLO, aligned with the rule updates.
- Developed the detection for other contest apparatuses using YOLO and trained the neural network.
- Deployed the network on NVIDIA embedded AI computers and accelerated its efficiency using TensorRT.

SKILLS

- **Programming Languages:** C/C++, Python, SQL
- **Software/Frameworks:** Linux (Arch Linux, Ubuntu), ROS, NVIDIA Isaac Sim and Lab, $\LaTeX$
- **Languages:** Chinese, English (ETS GRE: 331 (V: 161, Q: 170) + 5.0), French (DELF B2)